PROJECT REPORT

ELECTRIC VEHICLE MARKET ANALYSIS

Market Performance, Sales Trends & Battery Technology Insights

Project title	EV_Market_Analysis
Primary tool	Tableau Desktop
Workbook format	Tableau Packaged Workbook (.twbx)
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Institution / course	Medicaps University, BTech CSE
Submission date	2/11/26

Prepared as a project summary based on the workflow and Tableau work completed so far. Fill in the student and institution details before submission.

1. Executive Summary

This project explores the electric vehicle (EV) market through interactive data visualizations created in Tableau. The analysis focuses on sales performance across manufacturers and regions, year-wise sales trends, battery technology and battery-type patterns, and customer-rating information.

A Tableau dashboard was assembled from multiple worksheets, and a Tableau Story was created to present the dashboard as a guided view. A Manufacturer filter was tested successfully: selecting a manufacturer changed the relevant dashboard charts. A calculated field named **Sales per Vehicle** was also created in Tableau; its formula and business definition should be checked against the exact dataset fields before final submission.

The workbook was saved locally in Tableau Packaged Workbook format (.twbx). Tableau Public publishing and the Flask-based web embedding stage were identified as project requirements but are not yet confirmed as completed.

2. Project Goals and Scope

The project follows eight goals: (1) Data Collection and Extraction, (2) Data Preparation, (3) Data Visualization, (4) Dashboard, (5) Story, (6) Performance Testing, (7) Web Integration, and (8) Project Demonstration and Documentation.

The scope is to turn an EV dataset into a clear visual summary that helps viewers compare manufacturers, regions, model years, and battery-related categories. The report records the work completed so far and identifies the remaining validation tasks rather than assuming that every project requirement has already been finished.

3. Tools and Technologies

Tool / technology	Purpose
Tableau Desktop	Connecting the dataset, preparing worksheets, building visualizations, dashboard and Story.
CSV dataset	Source data for the EV market analysis. Exact filenames, row counts and source URL should be recorded from the original dataset.
Tableau Packaged Workbook (.twbx)	Local packaged project file for sharing or moving the workbook with supported data sources.
GitHub	Versioned project repository / portfolio location, if the workbook and supporting files are uploaded.
Python Flask (planned)	Web application framework intended for embedding the published Tableau dashboard and Story.

4. Methodology

4.1 Data Collection and Extraction

The EV dataset was downloaded and connected to Tableau. The exact source name, download date, file names, licence and original row counts should be copied from the dataset source or Tableau Data Source page before the report is submitted.

4.2 Data Preparation

The dataset was made available to Tableau for analysis. Before final submission, verify field names and data types, check for null or missing values, confirm that measures and dimensions are assigned appropriately, and document any cleaning or filtering actually performed. Do not claim that missing values were removed or transformed unless that operation was completed.

4.3 Data Visualization

Multiple worksheets were created to explore EV sales and related attributes. The project tracking notes refer to nine visualizations. The visible dashboard includes the following chart subjects:

- Top 10 Manufacturers by Sales
- Battery Technology by Sales
- EV Sales Trend by Model Year
- EV Sales by Country
- EV Sales by Market Region
- Average Battery Capacity by Type
- Battery Type Share by Vehicle Count
- Customer Rating vs. Vehicle Count
- One additional worksheet / visualization to be verified in the final workbook

Note: The ninth visualization should be identified from the final workbook. Confirm the exact chart titles and chart types in Tableau; do not use this provisional list as a substitute for checking the workbook.

4.4 Dashboard Design

The worksheets were arranged on a Tableau dashboard with a descriptive EV market title. The layout brings manufacturer sales, battery technology, model-year trends, country and regional sales, battery capacity, battery-type share and customer-rating information into one view. Review the dashboard at the intended laptop or presentation size to ensure titles, axes and labels remain readable and do not overlap.

4.5 Tableau Story

A Tableau Story was created and the dashboard was added as a story point. The Story title / workbook naming used in the project includes EV_Market_Analysis. The current number of story scenes should be checked in the Story tab; only one dashboard scene was explicitly described during the workflow.

5. Performance Testing

Performance testing checks whether the dashboard displays the intended data and responds correctly to user interaction. The following test log reflects the work reported so far.

Test item	What to verify	Current status
Amount of data rendered	Record count in Tableau and whether the expected data is visible; record source row count and any filters/exclusions.	Pending verification
Utilisation of filters	Select a Manufacturer value, observe relevant charts, then reset to All.	Manufacturer selection changed charts; reset and broader worksheet scope still to verify.
Calculated fields	Count calculated fields and validate each formula and its business meaning.	One field named Sales per Vehicle was created; formula needs validation.
Visualizations and graphs	Confirm the final number of worksheets / visualizations and check for blank or incorrect charts.	Nine visualizations were tracked; final count to verify.

Suggested test procedure

1. Record the exact row count from the Tableau Data Source page or source file.
2. Select one Manufacturer in the dashboard filter and note which worksheets respond.
3. Reset the filter to All and confirm the full view returns.
4. Check the calculated field formula against actual field names and expected units.
5. Count the final worksheets and dashboard views; inspect each for errors, blank marks or unreadable labels.

6. Web Integration

The planned web-integration work has two parts: publishing the Tableau dashboard and Story to Tableau Public, and embedding the published views in a Flask application. Saving a .twbx file locally does not publish it online. A Tableau Public profile login also does not, by itself, confirm that a workbook has been published.

Current status: Tableau Public publishing was not confirmed during the recorded workflow because the publishing option in the installed Tableau Desktop edition appeared unavailable. Flask embedding has not yet been completed.

Recommended next steps

• Confirm the exact Tableau edition and use a publishing route supported by that edition.
• Publish the workbook only after checking that the dataset contains no private, confidential or restricted information.
• Copy the published dashboard and Story URLs.
• Create a small Flask application and embed the published Tableau views using their official embed URLs or supported embedding method.
• Test the page in a browser and verify that the embedded views load and remain usable.

7. Project Demonstration and Documentation

For the final demonstration, explain the project objective, identify the dataset and fields used, show the key visualizations, demonstrate the Manufacturer filter, describe the Story, and summarize the insights that are directly supported by the charts. If the Flask integration is completed, finish by opening the web page and demonstrating the embedded dashboard and Story.

Suggested demonstration sequence

1. Introduce the EV market analysis objective.
2. Briefly describe the dataset and preparation steps actually performed.
3. Walk through the main visualizations and explain what each chart measures.
4. Demonstrate the Manufacturer filter and reset it to All.
5. Present the dashboard and Story.
6. Show the Tableau Public link and Flask page, if completed.
7. Close with key findings, limitations and possible future improvements.

8. Conclusion

The project has progressed through dataset download and Tableau connection, worksheet creation, dashboard assembly, Story creation and initial performance testing. The Manufacturer filter was tested and changed the displayed charts, and a calculated field was created. The workbook was saved in .twbx format. The remaining work is to verify the dataset and visualization counts, validate the calculated field, complete the required publishing and Flask embedding tasks, and prepare final evidence for demonstration and submission.

9. Final Submission Checklist

Item	Status to confirm
Original dataset source, licence and row counts documented	■
Data preparation steps and data types documented accurately	■
Final number and titles of visualizations verified	■
Dashboard layout checked at presentation size	■
Story scene count and captions verified	■
Manufacturer filter tested and reset to All	Partly completed
Amount of data rendered recorded	■
Calculated field formula validated and count recorded	■
TWBX workbook opens correctly on the target machine	■
Tableau Public dashboard and Story published; links recorded	■
Flask application embeds dashboard and Story	■
Screenshots, report and demo prepared	■

Important note for submission

This report is a project draft based on the information shared in the conversation. Before submitting it, fill in the student details, dataset source and exact row counts, confirm all chart titles and the ninth visualization, validate the calculated field, and update the Tableau Public / Flask status based on what

you actually complete.